

1. Where we are, and the current goal

What changed about the objective (read this first). The companion derivation (`new_approx_combined.txt`) builds $\text{score}_w(v)$ as a *lower bound* on $\|Av\|_2^2$ under assumptions (A1), (A2), and $\|N_w v\|_4 \leq \|Av\|_4$, using only data seen through window w (sketch B , current rows A_w , row sample R). The argmax of a lower bound is not, in general, the argmax of the bounded quantity. The present report changes the demand: in the experiments below, with $r = 2$, we now ask the score to *recover the top two right singular vectors* V_2 of A , i.e. to be argmax-correct, not just value-correct. We also relax the streaming scope to allow a future half-window A_{fut} . So before reading on, note the two shifts relative to the derivation:

- **Argmax-recovery, not lower-bounding.** M1 and M2 below are points where the lower-bound’s argmax disagrees with V_2 . M3 is the first repair signal for the single-window case where the argmax is far from the projected oracle. None of these is a bug in the inequality of `new_approx_combined.txt`; they are consequences of using a lower bound for a task it was not derived to solve.
- **Future rows allowed.** M3 and the replacement family S_6 introduce A_{fut} , the next half-window after the current half-window block A_w . Thus the sweep treats A_w itself as the current half-window and pairs it with the following same-size half-window. The original derivation excluded future rows. The experiments below add them because the recovery task needs more than past and current rows on some matrices.

Recap of the combined score. Specialising to the regime in which the earlier rows are measured at the scale of C_w (i.e. taking $\text{row}(N_w) \leftarrow \text{row}(C_w)$ in the base of the derivation’s correction factor, with $h_2(N_w v)/\log \text{row}(N_w)$ replaced by $h_2(C_w v)/\log \text{row}(C_w)$ in the exponent under (A2)), the combined score from the derivation can be written

$$\text{score}_w(v) = (\|Bv\|_2^2 + \|A_w v\|_2^2) \phi_w(v), \quad \phi_w(v) := \left(\frac{n}{\text{row}(C_w)} \right)^{h_2(C_w v)/(2 \log \text{row}(C_w))}, \quad (1)$$

where n is the total number of rows of A , $C_w = (R^\top, A_w^\top)^\top$ stacks the row sample R with A_w , $h_2(y) := 4 \log \|y\|_2 - 4 \log \|y\|_4$.

This paragraph introduces a diagnostic normalisation of the entropy multiplier used to discuss the claims below; it does not change the per-block measurements in Appendix A, which were collected with the implemented multiplier from the derivation’s notation at each block. Concretely, starting from the derivation’s $(\text{row}(N_w)/n)^{-h_2(C_w v)/(2 \log \text{row}(C_w))}$ and replacing $\text{row}(N_w)$ by $\text{row}(C_w)$ in that base gives $(\text{row}(C_w)/n)^{-h_2/(2 \log \text{row}(C_w))} = (n/\text{row}(C_w))^{h_2/(2 \log \text{row}(C_w))}$, the recap form. Equivalently, using the identity $h_2(y) = -\log(\|y\|_4^4/\|y\|_2^4)$, this

is the same object as the derivation’s $\|C_w v\|_4^{4c_w} \|C_w v\|_2^{-4c_w}$ once the substitution is applied; the $(n/\text{row}(C_w))^\cdot$ form is just the entropy multiplier rewritten so its base is ≥ 1 , with the sign of the exponent absorbing the sign of c_w . Since $h_2 \geq 0$ always and $n \geq \text{row}(C_w)$, the base $n/\text{row}(C_w) \geq 1$ and $\phi_w \geq 1$ in this recap form, in agreement with the derivation’s $(\text{row}(N_w)/n)^{-h_2/\dots}$ (base ≤ 1 , negative exponent). The M1, M2, and M3 measurements below do not depend on this substitution, and the empirical evidence in Appendix A is collected at intermediate blocks where $\text{row}(N_w) < n$ and $c_w < 0$ in the derivation’s notation.

Notation used by the diagnostics. Throughout this report, “oracle” (the top- r right singular vectors of A) is a global object defined from the full matrix; the score sees only window-local data. “Deflation” below means orthogonal projection of the search direction onto the orthogonal complement of the previously accepted $\hat{v}_j^{(\sigma)}$ ($j < i$): identical to the constraint $\hat{v}_j^{(\sigma)\top} v = 0$ in the derivation (`new_approx_combined.txt`, eq. for v_j). It does *not* mean deflating A itself. Likewise, V_{state} refers to the right factor V of the carried sketch $B = SV^\top$ from the derivation (the columns are $v_1, \dots, v_r$ stored at the start of the window); $\text{span}(V_{\text{state}}) = \text{rowspan}(B)$. For any window-local search that uses B , A_w , and A_{fut} , set

$$B_{\text{union}} := \text{rowspan} \begin{pmatrix} B \\ A_w \\ A_{\text{fut}} \end{pmatrix} \subset \mathbb{R}^d. \quad (2)$$

This is the search subspace available from those three matrices. Directions are extracted greedily: let $\hat{v}_i^{(\text{combined})}$ denote the i -th direction, defined by

$$\hat{v}_i^{(\text{combined})} \in \arg \max_{v \in \mathbb{S}^{d-1}, v \perp \hat{v}_j^{(\text{combined})} \forall j < i} \text{score}_w(v). \quad (3)$$

The “oracle” is the top- r right singular vectors V_r of A ; $\hat{v}_i^{(\text{oracle proj})}$ is the projection of the i -th oracle column v_i^* onto the feasible set actually used at slot i — i.e. onto the orthogonal complement of the previously accepted directions $\{\hat{v}_j^{(\sigma)}\}_{j < i}$ and, for a window-local search, also onto B_{union} . We compare against $\hat{v}_i^{(\text{oracle proj})}$ rather than v_i^* because no window-local score can possibly recover a direction outside its own search domain; the projected oracle is the strongest target the score is allowed to be measured against.

Two failures and one repair signal. Probing on a small set of test matrices identified two mechanisms by which the combined score loses $\hat{v}_2^{(\text{combined})}$ (and $\hat{v}_r^{(\text{combined})}$), plus one future-window signal that repairs the leading-direction single-window failure.

- **(M1) ϕ_w ranks the combined argmax above the oracle.** On mixed-tail-sharp at block 1, the score gap between $\hat{v}_2^{(\text{combined})}$ and $\hat{v}_2^{(\text{oracle proj})}$ sits entirely inside ϕ_1 (5.66 vs. 3.70, energy term flat at 0.819–0.833). See Appendix A.1. *Reconciliation with the derivation.* ϕ_w enters the lower bound

($\star$) via assumption (A2), which equates the (normalised) entropy of $C_w v$, $N_w v$, and Av . The inequality $\|Av\|_2^2 \gtrsim \|N_w v\|_2^2 \cdot (\text{row}(N_w)/n)^{-h_2(N_w v)/(2 \log \text{row}(N_w))}$ is correct as a value bound under (A2); M1 simply observes that *argmax*-ranking the bound is not the same as *argmax*-ranking $\|Av\|_2^2$. In the recap form $\phi_w = (n/\text{row}(C_w))^{h_2(C_w v)/(2 \log \text{row}(C_w))}$, larger ϕ_w means larger $h_2(C_w v)$, equivalently a *lower* ℓ_4/ℓ_2 concentration ratio. On this block the score-favoured direction has the larger ϕ_w and beats the oracle in the bound while contributing nothing extra to $\|Av\|_2^2$ globally.

- **(M2) The sketch term aligns $\hat{v}_2^{(\text{combined})}$ with $\text{span}(V_{\text{state}})$.** Once $\|Bv\|_2^2 \gg \|A_w v\|_2^2$ (empirically once $\rho_{\text{past}} := \|A_{1:w-1}\|_F^2 / \|A_w\|_F^2 \gtrsim 5$; this 5 is just where the energy ratio empirically tips on the test matrices, not a theorem), the energy term reduces to $\|Bv\|_2^2$, whose *argmax* over the deflation complement lies in $\text{span}(B)$. This is a directional dominance claim on the retained sketch subspace, not a claim that $\|B\|_F^2$ matches the full-history Frobenius norm; Appendix A.2 reports both scales separately. At the last measured block, block 31, on **mixed-tail-sharp**, $\text{state_align}(\hat{v}_2^{(\text{combined})}) = 0.989$ vs. $\text{state_align}(\hat{v}_2^{(\text{oracle proj})}) = 0.293$; a joint Stiefel-2 optimisation returns the same point. See Appendix A.2. *Reconciliation with the derivation.* `new_approx_combined.txt` justifies $B = \Sigma V_{\text{new}}^\top$ precisely because $\|Bv\|_2 = \|P_U M_w v\|_2$ on $\text{span}(V)$: the sketch is *designed* to make the energy term faithfully track the carried earlier-row sketch on the retained right span. M2 is therefore not a defect of B . It is the consequence of demanding *argmax*-recovery from a score whose energy term, by construction, tracks $\text{span}(V_{\text{state}})$ rather than $V_r(A)$ once ρ_{past} grows.
- **(M3) Add future rows to move the *argmax* toward the oracle projection.** At block 1 ($B = 0$) on **diffuse-diffuse**, $\hat{v}_1^{(\text{combined})}$ has $\cos^2 < 0.05$ against either exact oracle column and 0.096 against $\hat{v}_1^{(\text{oracle proj})}$, while an oracle-warm $\text{HM}_2(u_{\text{cur}}, u_{\text{fut}})$ isolation score reaches 0.999 against $\hat{v}_1^{(\text{oracle proj})}$. See Appendix A.3. Thus M3 is the repair signal for this failure: the current *argmax* is not near the projected oracle because the score sees only one half-window when $B = 0$. Adding A_{fut} lets the score test the stationarity condition used in `new_approx_future_hmean_online.txt`: the current and future per-row gains should agree for a real population direction. The harmonic mean then penalises a direction that is strong on A_w but weak on A_{fut} . Uniform-row-space matrices like **diffuse-diffuse** show why earlier rows plus current rows alone cannot distinguish population directions from draw-specific directions.

Each claim rests on per-block measurements; the detailed numbers are in Appendix A.

The common thread (working hypothesis). The score can prefer directions whose window-local evidence looks strong but is not oracle-identifying.

M1 and M2 are failure modes of the combined score. M3 is the corresponding fix direction for the single-window argmax failure. In M1 the score gap is in ϕ_w , which rewards the visible $h_2(C_w v)$ profile even when it does not correspond to extra global $\|Av\|_2^2$ energy. In M2 the energy term is dominated by $\|Bv\|_2^2$ once $\|Bv\|_2^2 \gg \|A_w v\|_2^2$, so the argmax tracks the leading directions of the carried sketch. M3 adds A_{fut} so a candidate direction must have support in the current and future half-windows, which moves the block-1 argmax back toward the projected oracle on **diffuse-diffuse**.

What the newer forward-replication diagnostics reject is only the narrower row-concentration version of this hypothesis. On the mixed-tail failure set, the score-favoured directions do *not* consistently have elevated current-window top-row share relative to the oracle, so the failure is not explained by a few rare rows in A_w creating a spike that then disappears on A_{fut} . The verified signatures are instead current-future energy imbalance ($u_{\text{cur}} \gg u_{\text{fut}}$ for the score-favoured direction while the oracle keeps $u_{\text{cur}} \approx u_{\text{fut}}$) and carry pinning (state_align near one for the score-favoured direction). Thus the failure is unstable visible evidence, not current-window row spikes alone.

The replacement family S_6 . Status note. Unlike score_w , the family S_6 below is *not* the maximiser of any inequality on $\|Av\|_2^2$; it has no analogue of (A1)/(A2) and no lower-bound theorem behind it. It is a heuristic engineered to be argmax-correct on representative test matrices. The reader should not try to map $u_{\text{sk}}, u_{\text{cur}}, u_{\text{fut}}$ back onto $\|Bv\|_2^2 + \|A_w v\|_2^2$ and ϕ_w : we are no longer in the lower-bound framework. M1 and M2 motivate leaving the combined score; M3 motivates adding the future-window stationarity test. Together they led to a value-only, analytically differentiable family that (i) sees a future half-window A_{fut} from after the current one, (ii) avoids entropic gating, and (iii) uses a rank-aware sketch term that does not collapse to $\text{span}(V_{\text{state}})$. This was the best previous operational replacement before the frame-level sufficiency sweep in §2-§3; it is a baseline under diagnosis here, not a surviving recommendation. The row sample R used in the derivation’s $C_w = (R^\top, A_w^\top)^\top$ is intentionally not a slot in these replacement scores: the replacement line discards row-sample entropy evidence and substitutes current-future replication evidence instead. The score is

$$S_6(v) := \text{HM}_3(u_{\text{sk}}(v), u_{\text{cur}}(v), u_{\text{fut}}(v)), \quad (4)$$

where each slot is a normalised Rayleigh quotient

$$u_{\text{sk}}(v) := \frac{\|Bv\|_2^2}{u_{\text{sk}}}, \quad u_{\text{cur}}(v) := \frac{\|A_w v\|_2^2}{\|A_w\|_F^2}, \quad u_{\text{fut}}(v) := \frac{\|A_{\text{fut}} v\|_2^2}{\|A_{\text{fut}}\|_F^2}, \quad (5)$$

and HM_3 is the harmonic mean $3/(u_{\text{sk}}^{-1} + u_{\text{cur}}^{-1} + u_{\text{fut}}^{-1})$. Greedy extraction yields $\hat{v}_i^{(S_6)}$ analogously to $\hat{v}_i^{(\text{combined})}$. The harmonic mean’s smallest-link enforcer rewards directions that *simultaneously* have non-trivial $\|Bv\|_2^2$, $\|A_w v\|_2^2$, and $\|A_{\text{fut}} v\|_2^2$ in the sketch, current, and future windows — ruling out A_w -only and

sketch-only directions in one shot. The weight w_{sk} is the rank- r truncation of $\|B\|_F^2$, i.e. $w_{\text{sk}} := \sum_{j=1}^r s_j^2$ where $s_1 \geq \dots \geq s_r$ are the singular values stored in the carried sketch $B = SV^\top$; this is the normalising constant that places each u_X in $[0, \sim 1]$.

The recent line of work: balancing HM_3 , and why it hit a wall. HM_3 's smallest-link property only fires correctly if the oracle direction itself has roughly balanced $(u_{\text{sk}}, u_{\text{cur}}, u_{\text{fut}})$. The oracle is far from balanced where S_6 fails (the ratio $u_{\text{sk}}/u_{\text{cur}}$ at the last measured block, block 31, runs from $1.74\times$ to $412\times$ across the five matrices listed in Appendix A.4's slot-ratio table), so we tried a per-direction reweighting — the “ E_2 weighting” — that, inside $u_{\text{sk}}(v) = \|Bv\|_2^2/w_{\text{sk}}$ only (the u_{cur} and u_{fut} slots are unchanged), replaces the scalar w_{sk} with $\sigma_{\ell(v)}^2$ for $\ell(v) := \arg \max_{j \leq r} (v_j^\top v)^2$ the sketch-direction index most aligned with v (here v_j are the columns of V_{state} and $\sigma_j = s_j$ the corresponding stored singular values). The audit metric reports success (E_2 drives the oracle's max/min ratio across the three slots $(u_{\text{sk}}, u_{\text{cur}}, u_{\text{fut}})$ inside $2.4\times$ on the regression matrices — the three matrices on which the streaming benchmark regressed under S_6 , namely **diffuse-diffuse**, **residual-spiky-shocks**, and **mixed-tail-soft**; this trio is the working set for g1 's landscape probe and Appendix A.4) but the streaming benchmark *regresses* sharply on the same matrices, and a landscape probe at the last measured block, block 31, finds the cause: under E_2 , the score's argmax sits strictly above the score at the oracle by $+0.041$, $+0.047$, $+0.006$ on **diffuse-diffuse**, **residual-spiky-shocks**, **mixed-tail-soft**. The gap survives a high-budget joint sweep (200 random restarts, $\text{maxit} = 2000$, $\hat{v}_1^{(S_6)}$ anchored at $\hat{v}_1^{(\text{oracle proj})}$, gap stays $\geq +0.122$ on **diffuse-diffuse**), ruling out under-optimisation. The conclusion is structural: per-direction reweighting balances the oracle's slot values but relocates the score's actual peak away from the oracle. No further scalar weight tuning fixes this. Detailed numbers, including the seven-matrix capture table under S_6 , E_2 , **force_oracle_v2** (abbreviated **force_02** in the appendix table; replaces only $\hat{v}_2^{(S_6)}$ with $\hat{v}_2^{(\text{oracle proj})}$ at every block), and **force_oracle_frame** (abbreviated **force_frame**; replaces both directions with the oracle rank-2 frame), are in Appendix A.4.

The singular goal. The remaining question collapses to one:

“what minimal information, computable from window-local data, makes the global oracle subspace the unique maximising subspace of the score against every adversarial combination of component-local tails?”

Formally, for the rank-2 tests in this report: fix a feature set $\mathcal{E}$ available at window w . In this report, *deployable window-local* features are computed from B , the current block A_w , the next half-window A_{fut} , and products of those same matrices. An explicit $A^\top A$ or V_r term is a *diagnostic global* feature: it may be used as a ceiling or sanity check, but not as an operational score. Approximations to $A^\top A$ are only admissible after they are constructed from

window-local data and pass the same sufficiency test as the value-only features. Given such an $\mathcal{E}$, define a frame-level score $S_{\mathcal{E}}(Z)$ on pairs of orthonormal directions $Z = [z_1, z_2] \in \text{St}(d, 2)$, and maximise it inside the already defined search subspace B_{union} . Let Q_{union} denote any orthonormal basis matrix for this subspace, so $\text{range}(Q_{\text{union}}) = B_{\text{union}}$, and let $P_{\text{union}} = Q_{\text{union}}Q_{\text{union}}^{\top}$ be the orthogonal projector onto B_{union} . The projected oracle frame Z_{oracle} is obtained by taking a thin QR (equivalently, the polar factor) of $P_{\text{union}}V_2$; if $\text{rank}(P_{\text{union}}V_2) < 2$, the missing direction is unreachable and the feature set fails the recovery test before optimisation. With this definition, the test is

$$Z_{\text{oracle}} \stackrel{?}{=} \arg \max_{Z \in \text{St}(d, 2), \text{range}(Z) \subset B_{\text{union}}} S_{\mathcal{E}}(Z). \quad (6)$$

For right-rotation-invariant scores such as $\|A_X Z\|_F^2$, equality means equality of the recovered two-plane; oriented-frame equality is only meaningful for scores that break the $Z \mapsto ZQ$ symmetry. If the projected oracle subspace is the maximiser, $\mathcal{E}$ is sufficient and an operational score can be built around it. Otherwise no slot reweighting recovers it and $\mathcal{E}$ must change.

Why this direction (H3) among possible solutions. Three broad directions are open for addressing the misleading-tail problem; we choose the third.

- **H1: keep $\{u_{\text{sk}}, u_{\text{cur}}, u_{\text{fut}}\}$, design a smarter aggregator or weighting.** The line that produced E_2 and the broader balance-tuning family. The AB-03 closure (the high-budget joint sweep summarised in Appendix A.4; “AB-03” is the internal experiment ID for the E_2 -vs- S_6 landscape probe) refutes it as a controlling principle: balance-tuning fails structurally ($\Delta_{S_6+E_2} \geq +0.122$ even at high optimisation budget with the oracle anchor) because the feature set itself does not distinguish the oracle from adversarial component-tail combinations. Reweighting cannot create information the features do not carry.
- **H2: replace value-only features with an explicit global quantity (e.g. $A^{\top}A$ or its leading invariant subspace).** Closes the gap by construction — $A^{\top}A$ ’s leading invariant subspace *is* the oracle — but abandons the streaming, window-local, value-only constraints that motivated the score family. Worth investigating eventually, but not before the cheaper options are exhausted.
- **H3 (chosen): treat the problem as one of evidence sufficiency, sweep candidate features in increasing intrusiveness, and use a frame-level audit to decide each one.** H3 sits between H1 and H2: it preserves the value-only, streaming-compatible setting but stops reweighting a fixed feature set; instead, for each candidate addition $\Delta\mathcal{E}$, it asks whether Z_{oracle} becomes the maximiser on $\text{St}(d, 2) \cap B_{\text{union}}$ under joint optimisation. Three properties of H3 motivate the choice:

1. *Per-feature, falsifiable output.* A feature either makes the oracle the argmax (the test passes) or it does not (the test fails). There is no tuning loop and no audit metric whose correlation with bench has to be assumed.
2. *Separation of score-design failure from optimiser failure.* Greedy and joint Stiefel ascent are run side by side ($j - g$ in Appendix B); when $j - g$ is small both routines reach the same maximum, so any residual $\Delta_{\text{joint}} > 0$ is a property of the feature set, not the search.
3. *Reusable screens.* The canonical screens used in H3 (oracle-vs-winner gap at vector and frame level, force-oracle ceilings, anchor sensitivity, and the existing vector-level row-cheat dominance screen — checking that a vector score does not collapse onto a direction that merely picks out a single high-norm row of A_w or A_{fut}) are also the acceptance gates that any future H1- or H2-style proposal must pass. The analogous frame-level row-cheat screen is still open work.

H3 also preserves the value-only, streaming, window-local constraints that motivated the score family, and it requires no new infrastructure beyond the Stiefel ascent already built for S_6 .

2. Method: a sequential evidence-sufficiency sweep

Lifting to frames. Throughout this section, $X \in \{\text{sk}, \text{cur}, \text{fut}\}$ indexes the three sources, with $A_{\text{sk}} := B$, $A_{\text{cur}} := A_w$ the current half-window block, and A_{fut} the following same-size half-window. Extend each u_X to an orthonormal pair $Z = [z_1, z_2] \in \text{St}(d, 2)$ via

$$u_X(Z) := \frac{\|A_X Z\|_F^2}{\|A_X\|_F^2} = \frac{\|A_X z_1\|_2^2 + \|A_X z_2\|_2^2}{\|A_X\|_F^2}. \quad (7)$$

A small bookkeeping change relative to the §1 vector definitions: the sketch slot at the frame level uses the full $\|B\|_F^2$ as denominator for symmetry with $\|A_w\|_F^2$ and $\|A_{\text{fut}}\|_F^2$, whereas the vector-level $u_{\text{sk}}(v)$ in §1 uses the rank- r truncation $w_{\text{sk}} = \sum_{j=1}^r s_j^2$. Since the carried sketch $B = SV^\top$ has exactly r stored singular values, $\|B\|_F^2 = \sum_{j=1}^r s_j^2 = w_{\text{sk}}$ *identically* in this setting; the two normalisations agree. We retain the $\|B\|_F^2$ notation at the frame level for symmetry with the other two slots. Even were the two constants to differ, they would scale u_{sk} uniformly in v and so leave every argmax inside a fixed window unchanged. This is invariant under $Z \mapsto ZQ$, $Q \in O(2)$, hence depends only on $\text{span}(Z)$. The base frame score is

$$S_{\text{HM}}(Z) := \text{HM}_3(u_{\text{sk}}(Z), u_{\text{cur}}(Z), u_{\text{fut}}(Z)). \quad (8)$$

The four feature sets we tried, in order. We test increasingly intrusive feature sets and ask whether each makes $\hat{Z}^{(\sigma)} = Z_{\text{oracle}}$. Mapping back to the

H1/H2/H3 taxonomy of §1: V1 and V2 are H3-style $\Delta\mathcal{E}$ additions (a new A_{fut} -only feature; a new current–future cross feature); V3 is an optimisation-control baseline using the original H1 feature set $\{u_{\text{sk}}, u_{\text{cur}}, u_{\text{fut}}\}$ with no aggregator or weighting change, so it isolates the joint-optimisation question from the feature question; V4 is the H2 ceiling-diagnostic (oracle information injected explicitly to upper-bound how much closure any window-local feature could buy).

V1. HM with a future-direction term.

$$S_1(Z) := S_{\text{HM}}(Z) \cdot \frac{\|A_{\text{fut}}^\top A_{\text{fut}} Z\|_F^2}{\|A_{\text{fut}} Z\|_F^2}. \quad (9)$$

Multiplies the harmonic mean by a Rayleigh-style future-direction strength, weighted by $A_{\text{fut}} Z$. Tests whether Z_{oracle} aligns with the leading singular directions of A_{fut} alone.

V2. HM with a current-future cross term.

$$S_2(Z) := S_{\text{HM}}(Z) \cdot \frac{\langle A_{\text{cur}} Z, A_{\text{fut}} Z \rangle_F}{\|A_{\text{cur}} Z\|_F \|A_{\text{fut}} Z\|_F}. \quad (10)$$

This diagnostic uses the same-size-half-window convention: A_{cur} and A_{fut} have the same row count, and the Frobenius inner product pairs row i of the current half-window with row i of the future half-window. It boosts subspaces along which $A_{\text{cur}} Z$ and $A_{\text{fut}} Z$ are aligned in $\mathbb{R}^{n_w \times 2}$ under that row-paired convention. The raw cosine factor is allowed to be negative; no clipping is applied. In the reported windows the denominators are nonzero, and a zero denominator would make the candidate frame inadmissible for this diagnostic. Tests whether Z_{oracle} sits where current and future windows agree directionally.

V3. HM, joint Stiefel optimization.

$$S_3(Z) := S_{\text{HM}}(Z), \quad \text{maximised jointly over } Z \in \text{St}(d, 2). \quad (11)$$

Same feature set, but replaces the sequential “rank-1, deflate, rank-1” with a joint 2-frame ascent. Tests whether the per-direction decomposition was the bottleneck.

V4. HM plus an oracle subspace reward.

$$S_4(Z) := S_{\text{HM}}(Z) + \frac{\lambda}{2} \|V_r^\top Z\|_F^2. \quad (12)$$

Adds a term rewarding Z for lying in $\text{span}(V_r)$. Uses information *not* available from B , A_w , A_{fut} alone—a diagnostic, not a deployable score. Quantifies how much oracle information must be injected before $\hat{Z}^{(S_4)} = Z_{\text{oracle}}$.

Optimizer. For each of V1–V4 we maximise $S(Z)$ over 2-frames Z whose range lies in B_{union} . Write $Z = Q_{\text{union}}W$ with $W \in \text{St}(m, 2)$, where $m = \dim B_{\text{union}}$. The Euclidean gradient of every u_X is closed-form,

$$\nabla_Z u_X(Z) = \frac{2A_X^\top A_X Z}{\|A_X\|_F^2}, \quad (13)$$

so ∇S_{HM} is closed-form by the chain rule on HM_3 , and likewise for S_1, S_2, S_4 . Pull the gradient back to W and run *polar retraction* on $\text{St}(m, 2)$:

$$W^+ \leftarrow \text{polar}(W + \eta(I - WW^\top)G_W), \quad (14)$$

where G_W is the Euclidean gradient with respect to W and η is set by back-tracking line search on the score. Random restarts are warm-started from (i) the feasible projected oracle $Z_{\text{oracle}} = \text{polar}(P_{\text{union}}V_2)$, (ii) the projection of the streaming state’s leading directions into B_{union} , and (iii) uniform Stiefel samples in the Q_{union} coordinates. For each matrix and each feature set we report two numbers side by side:

- *Greedy*: per-direction procedure (rank-1 ascent, deflate, rank-1 ascent on the orthogonal complement), yielding $\hat{v}_1^{(\sigma)}, \hat{v}_2^{(\sigma)}$;
- *Joint*: the joint $\text{St}(d, 2)$ ascent above, yielding $\hat{Z}^{(\sigma)}$.

The difference $j - g$ is the optimization audit: large means the per-direction decomposition left score on the table; near zero means both routines reach the same maximum and the gap to Z_{oracle} is a property of the score, not the optimizer.

3. Evidence and conclusion

Headline outcome. On diffuse-diffuse, residual-spiky-shocks, and mixed-tail-soft at block 31, the four feature sets close as follows:

- **V1 (HM with future-direction term): fail.** The A_{fut} direction term changes both $S_1(Z_{\text{oracle}})$ and the score at the non-oracle argmax, so Δ_{joint} *shrinks* numerically (compare V1 and V3 rows in Appendix B.1) without making the oracle the maximiser. Because V1 multiplies HM by a squared-energy scale, the V1 and V3 score values should be compared as within-feature oracle-vs-winner gaps, not as absolute cross-feature scores. The oracle does not align strongly enough with the leading singular directions of A_{fut} alone.
- **V2 (HM with current-future cross term): fail harder.** $S_2(Z_{\text{oracle}}) \approx 0$ on two of three matrices, so the oracle is actively *anti-selected*: Z_{oracle} is not where $A_w Z$ and $A_{\text{fut}} Z$ agree directionally under the normalised Frobenius inner product. “Fail harder” here means the feature is not merely uninformative but pathological — it ranks Z_{oracle} near the bottom rather than near the top.

- **V3 (joint Stiefel ascent on plain HM): fail.** Joint matches greedy to within $|j-g| \leq +0.0004$, so per-direction decomposition is not the bottleneck. $\hat{Z}^{(S_3)}$ remains strictly above Z_{oracle} by $\Delta_{\text{joint}} \in [+0.014, +0.032]$.
- **V4 (HM plus $\lambda\|V_r^\top Z\|_F^2$): partial closure at the span level.** On residual-spiky-shocks and mixed-tail-soft, $\lambda = 0.1$ pushes both squared principal cosines into $[0.85, 0.99]$ (specifically $\text{pa}^2[0] \in \{0.99, 0.99\}$ and $\text{pa}^2[1] \in \{0.85, 0.93\}$, see Appendix B.2); **diffuse-diffuse** needs $\lambda = 0.5$. A residual gap ~ 0.003 – 0.005 survives at $\lambda = 0.5$ because $\|V_r^\top Z\|_F^2$ is invariant under right-rotations of Z inside $\text{span}(V_r) \cap B_{\text{union}}$.

The full V1–V3 score table and the V4 λ -sweep with principal-angle data are in Appendix B.

Implication. The feature set $\{u_{\text{sk}}, u_{\text{cur}}, u_{\text{fut}}\}$ —even augmented with rotation-invariant terms built from A_{fut} alone or from a current-future cross product—is *insufficient at rank 2 to identify* $\text{span}(V_2)$. Z_{oracle} is defined by global geometry of $A^\top A$, while the available features are window-local energies. Closing the gap requires either

- (a) actual oracle-direction information (V4-style; diagnostic only, not deployable), or
- (b) a window-local quantity approximating the leading invariant subspace of $A^\top A$ from streaming data—e.g. a power-iteration-style inner product using stacked current-and-future rows in place of $A^\top A$. Any such quantity must itself pass the same evidence-sufficiency test (V3-style joint sweep) before entering the operational score.

The cheaper feature additions have been ruled out.

Status. The closure is now the operating principle for any future change to the combined score: any new feature must first pass a joint $\text{St}(d, 2)$ ascent against the oracle frame Z_{oracle} , with Z_{oracle} as the argmax, before any operational benchmark. Open work, in priority order:

1. Reconcile a small numerical discrepancy in the oracle-frame score on **diffuse-diffuse** between an earlier landscape probe (score at $\hat{v}_2^{(\text{oracle proj})}$ logged at ~ 0.297 , not reproduced in the tables of this report) and the high-budget probe in Appendix A.4 ($\sigma(\hat{v}_2^{(\text{oracle proj})}) = 0.2164$ under $S_6 + E_2$). Likely a streaming-state seeding difference; pin the RNG.
2. Build a frame-level row-cheat diagnostic to complement V4.
3. Investigate window-local approximations of the leading invariant subspace of $A^\top A$, each gated by its own V3-style sufficiency test before any aggregator change.

Appendix A. Evidence for the combined-score failures and the future-window repair signal

A.1 (M1) The entropy term ϕ_w penalises the oracle

Setup. **mixed-tail-sharp**, block 1 ($B = 0$, isolating ϕ_w from any sketch effect). Restrict to the deflation complement of $\hat{v}_1^{(\text{combined})}$. Inside this complement $\|A_w v\|_2^2$ is essentially flat (0.819–0.833), so score variation is driven entirely by ϕ_1 .

Measurements. For two named directions and five random controls we record ϕ_1 , the score, and squared alignments with the oracle directions $\hat{v}_2^{(\text{oracle proj})}$ and v_1^* (the latter confirms we are testing the second slot):

direction	ϕ_1	score ₁	align _{v_2^*}	align _{v_1^*}
$\hat{v}_2^{(\text{combined})}$ (score argmax)	5.66	4.68	0.016	$\leq 10^{-3}$
$\hat{v}_2^{(\text{oracle proj})}$	3.70	3.08	0.749	$\leq 10^{-3}$
random direction 0	3.11	2.57	0.022	$\leq 10^{-3}$
random direction 1	3.43	2.83	0.038	$\leq 10^{-3}$
random direction 2	3.62	2.99	0.012	$\leq 10^{-3}$
random direction 3	3.89	3.21	0.008	$\leq 10^{-3}$
random direction 4	3.04	2.51	0.031	$\leq 10^{-3}$

Reading. $\hat{v}_2^{(\text{combined})}$ has $\phi_1 = 5.66$, $\hat{v}_2^{(\text{oracle proj})}$ has $\phi_1 = 3.70$, and every random direction sits in $\phi_1 \in [3.0, 3.9]$. The score gap $4.68 - 3.08 = 1.60$ is contained entirely in the ϕ_1 multiplier; the energy term cancels by construction. With $B = 0$, the gap does not involve any sketch term, so ϕ_w alone is responsible. The ranking $\phi_1(\hat{v}_2^{(\text{combined})}) > \phi_1(\hat{v}_2^{(\text{oracle proj})})$ on this matrix is the direct measurement of M1.

A.2 (M2) The sketch term aligns $\hat{v}_2^{(\text{combined})}$ with $\text{span}(V_{\text{state}})$

Setup. Same matrix **mixed-tail-sharp**, blocks $b \in \{1, 6, 12, 31\}$; block 31 is the last measured block. Define $\text{state_align}(v) := \|V_{\text{state}}^\top v\|_2^2$ and evaluate it at $\hat{v}_2^{(\text{combined})}$ and at $\hat{v}_2^{(\text{oracle proj})}$. DIAG-05 (the per-block diagnostic probe on **mixed-tail-sharp**) logs the full-history scale $\rho_{\text{past}} := \|A_{1:w-1}\|_F^2 / \|A_w\|_F^2$ and the sketch scale $\rho_B := \|B\|_F^2 / \|A_w\|_F^2$; ρ_{past} is the “ $\rho \gtrsim 5$ ” sketch-dominance scale used in the original note, while ρ_B remains small because B stores only the retained rank-2 action.

b	ρ_{past}	ρ_B	state_align($\hat{v}_2^{(\text{combined})}$)	state_align($\hat{v}_2^{(\text{oracle proj})}$)	state_align($\hat{v}_2^{(\text{combined, joint})}$)
1	0	0	—	—	—
6	5.006	0.063	0.837	0.167	0.834
12	11.011	0.063	0.882	0.297	0.897
31	30.275	0.071	0.988	0.293	0.988

Reading. Once $\rho_{\text{past}} \gtrsim 5$, the energy term reduces to $\|Bv\|_2^2$, whose argmax over the deflation complement of $\hat{v}_1^{(\text{combined})}$ lies in $\text{span}(B) = \text{span}(V_{\text{state}})$. At the last measured block, block 31, $\text{state_align}(\hat{v}_2^{(\text{combined})}) = 0.988$ while $\text{state_align}(\hat{v}_2^{(\text{oracle proj})}) = 0.293$. The older appendix note recorded 0.103 for the oracle; DIAG-05 attributes the difference to a current-code projection/deflation detail.

Joint Stiefel-2 control. Replacing the greedy rank-1, deflate, rank-1 procedure with a joint Stiefel-2 optimisation under the same combined score gives the tabled joint state alignments 0.834, 0.897, and 0.988 at $b = 6, 12, 31$, respectively, with alignments to v_2^* matching the greedy run to the reported precision. At block 1 it nudges the alignment to 0.247 but at strictly lower score — a different stationary point, not a higher maximum. The failure of the second direction is therefore intrinsic to the score on $\text{St}(d, r)$, not an artefact of the deflation order.

A.3 (M3) Adding A_{fut} moves the argmax toward the oracle projection

Setup. diffuse-diffuse, where the population row-space is uniform: a single half-window’s top right singular vector is a draw-specific noise direction rather than a population direction. We compare four scores at block 1 ($B = 0$, isolating the effect of not using A_{fut} from M1/M2) and across the streaming benchmark to the last measured block, block 31. The block-1 row labelled “ $\text{HM}_2(u_{\text{cur}}, u_{\text{fut}})$ ” is the two-slot variant of S_6 used for this M3-isolation experiment (u_{sk} is dropped because $B = 0$); the streaming-bench “ S_6 ” row below uses the full HM_3 form from §1.

*Block-1 alignments to $\hat{v}_1^{(\text{oracle proj})}$ on *diffuse-diffuse*.*

score	$\cos^2(\hat{v}_1, \hat{v}_1^{(\text{oracle proj})})$	reading
combined (no A_{fut})	0.096	u_{fut} nearly zero
$\text{HM}_2(u_{\text{cur}}, u_{\text{fut}})$, oracle-warm	0.999	balanced current/future image

Streaming-bench, terminal $\cos_1^2$ at the last measured block, block 31, same matrix. Here $\cos_1^2$ is the one-based shorthand for the leading direction’s squared oracle alignment; Appendix A.4 later uses zero-based slot indices for a two-direction capture sum.

method	terminal $\cos_1^2$
combined	0.030
S_6	0.005
incremental SVD	0.239
future_hmean_online (oracle-aware ceiling)	0.528

Reading. At block 1 with $B = 0$, the combined score maximises $\|A_w v\|_2^2$ alone; on a uniform-row-space matrix this selects the top right singular vector

of a single half-window, a draw-specific direction far from the projected oracle. Adding A_{fut} gives the score a persistence test across two halves. HM_2 across both halves then recovers the population direction (0.999) because the harmonic mean rewards directions with non-trivial $\|A_X v\|_2^2$ in both halves. M3 is therefore the repair signal for the leading-direction failure on uniform-row-space matrices: on **mixed-tail-sharp**, where the population top direction is dominant, $\hat{v}_1^{(\text{combined})}$ has $\cos_0^2 = 0.891$ in the current DIAG-05 single-matrix streaming control (older bench note: 0.785); M1 and M2 explain the second-direction failure there, so the future-row repair is not the main issue for that case.

A.4 Why HM_3 balance-tuning hit a wall

Oracle slot ratio across matrices. The ratio $u_{\text{sk}}/u_{\text{cur}}$ at v_1^* at block 31, taken from a per-block component-time-series probe:

matrix	$u_{\text{sk}}/u_{\text{cur}}$ at $b=6$	at $b=31$	S_6 terminal $\cos_1^2$
etf-basket-basis	1.36×	1.74×	0.652
mixed-tail-balanced	2.49×	0.85×	0.0004
mixed-tail-sharp	3.10×	0.60×	0.013
diffuse-diffuse	0.70×	21.5×	0.005
static-cex	72×	412×	0.025

This table shows **etf-basket-basis** as the only matrix where the displayed sketch/current oracle ratio is near balance, and it is the only matrix where S_6 partially works. The matrices on which S_6 fails worst exhibit the largest displayed sketch/current oracle imbalance; the full max/min slot-ratio audit used for E_2 is summarised in §1.

Landscape probe under the E_2 reweighting. Compare the score at its own argmax against the score at $\hat{v}_2^{(\text{oracle proj})}$ at block 31, across three regression matrices. Δ is reported for both S_6 (baseline) and S_6+E_2 :

matrix	Δ_{S_6}	$\Delta_{S_6+E_2}$
diffuse-diffuse	+0.007	+0.041
residual-spiky-shocks	+0.007	+0.047
mixed-tail-soft	+0.004	+0.006

Under baseline S_6 , the winner is effectively tied with the oracle at this vector-level probe (gap ≤ 0.007), but still not a strict oracle win; under S_6+E_2 , the argmax sits strictly above the oracle.

*High-budget joint sweep on **diffuse-diffuse** at $b=31$.* 200 random restarts, maxit = 2000, tol = 10^{-12} ; second-direction ascent under three choices of the leading direction $\hat{v}_1^{(\sigma)}$:

$\hat{v}_1^{(\sigma)}$	score σ	$\sigma(\hat{v}_2^{(\sigma)})$	$\sigma(\hat{v}_2^{(\text{oracle proj})})$	Δ
high-budget winner of S_6+E_2 over B_{union}	S_6	0.0190	0.0188	+0.0001
high-budget winner of S_6+E_2 over B_{union}	S_6+E_2	0.3390	0.2164	+0.1226
$\hat{v}_1^{(\text{oracle proj})}$	S_6	0.0189	0.0188	+0.0002
$\hat{v}_1^{(\text{oracle proj})}$	S_6+E_2	0.3387	0.2164	+0.1223
$V_{\text{state}}[:, 0]$	S_6	0.0189	0.0188	+0.0002
$V_{\text{state}}[:, 0]$	S_6+E_2	0.3398	0.2164	+0.1235

Anchoring the leading direction at v_1^* does not help: the gap holds at +0.122. The gap is structural, not a greedy-anchoring artefact.

Force-oracle ceiling probe (seven matrices, capture = $\cos_0^2 + \cos_1^2$ where $\cos_i^2 := (\hat{v}_{i+1}^{(\sigma)\top} v_{i+1}^)^2$ is the squared alignment of the $(i+1)$ -th extracted direction with the $(i+1)$ -th oracle column — the 0,1 indices here are zero-based slot indices, distinct from the $\cos_1^2$ first-direction shorthand used in the $\text{g}1$ streaming-bench table; max 2.00, half-window 64).* “ S_6 cap” is the operational S_6 score-function pipeline; “ E_2 cap” is S_6+E_2 ; “**force_02** cap” replaces only $\hat{v}_2^{(S_6)}$ with $\hat{v}_2^{(\text{oracle proj})}$ at every block; “**force_frame** cap” replaces both directions with the rank-2 SVD frame of $[v_1^*, v_2^*]$:

matrix	S_6 cap	E_2 cap	force_02 cap	force_frame cap
diffuse-diffuse	1.05	0.66	1.00	1.95
residual-spiky-shocks	1.18	1.02	1.01	1.92
mixed-tail-soft	1.08	1.55	1.00	1.95
mixed-tail-sharp	1.07	1.09	1.00	1.95
static-cex	1.12	1.05	1.00	1.95
mixed-tail-balanced	1.09	1.19	1.00	1.95
etf-basket-basis	1.64	1.54	2.00	2.00

Two readings. (i) The $\cos^2$ ceiling is $\approx 1.95/2.00$ on every matrix — $\text{rowspan}(B_{\text{union}})$ contains both oracle directions, so the bench is nowhere near its information-theoretic ceiling and the operational regression is not a reachability problem. (ii) Forcing only $\hat{v}_2^{(S_6)} \mapsto \hat{v}_2^{(\text{oracle proj})}$ recovers ≈ 1.00 of the 1.95 ceiling on the six regression-prone matrices: $V_{\text{default}}[:, 0]$ (the leading direction produced by the unmodified operational S_6 pipeline, i.e. $\hat{v}_1^{(S_6)}$) from streaming is itself a poor approximation of v_1^* on those matrices, so the rank-2 frame $[V_{\text{default}}[:, 0], v_2^*]$ catches one oracle direction only. The leading direction is also broken; replacing the second alone is insufficient. The lone exception is **etf-basket-basis**, where **force_02** reaches 2.00 (matching **force_frame**): on this matrix $\hat{v}_1^{(S_6)} \approx v_1^*$ already, so replacing the second direction alone closes both slots. This is consistent with $\text{g}1$ ’s observation that **etf-basket-basis** is the only matrix on which S_6 partially works (oracle slots are roughly balanced there).

Appendix B. The V1–V4 sweep, in full

All numbers are at block 31. “oracle” is the score at Z_{oracle} , “greedy” is the score at $\hat{Z}_{\text{greedy}}^{(\sigma)}$ (slot-by-slot ascent), “joint” is the score at $\hat{Z}_{\text{joint}}^{(\sigma)}$ (Stiefel- $(d, 2)$ ascent), $\Delta_{\text{joint}} = \text{joint} - \text{oracle}$, and $j - g = \text{joint} - \text{greedy}$ is the optimisation audit.

B.1 V1, V2, V3 (no oracle term)

feature set	matrix	oracle	greedy	joint	Δ_{joint}	$j - g$
HM (V3)	diffuse-diffuse	0.0066	0.0380	0.0383	+0.0316	+0.0003
HM (V3)	residual-spiky-shocks	0.0270	0.0409	0.0409	+0.0140	+0.0001
HM (V3)	mixed-tail-soft	0.0215	0.0382	0.0386	+0.0170	+0.0004
HM $\times$ futdir (V1)	diffuse-diffuse	0.0099	0.0315	0.0318	+0.0219	+0.0002
HM $\times$ futdir (V1)	residual-spiky-shocks	0.0188	0.0278	0.0279	+0.0091	+0.0001
HM $\times$ futdir (V1)	mixed-tail-soft	0.0223	0.0322	0.0324	+0.0101	+0.0002
HM $\times$ crosscorr (V2)	diffuse-diffuse	0.0128	0.0374	0.0376	+0.0248	+0.0002
HM $\times$ crosscorr (V2)	residual-spiky-shocks	0.0000	0.0392	0.0393	+0.0393	+0.0001
HM $\times$ crosscorr (V2)	mixed-tail-soft	0.0008	0.0381	0.0383	+0.0375	+0.0002

Three readings, each load-bearing.

1. $|j - g| \leq +0.0004$ uniformly: the joint optimiser matches the greedy one to within tolerance, so the slot-by-slot decomposition is not the bottleneck and the optimiser is well-tuned.
2. $\Delta_{\text{joint}} > 0$ on every cell: the score’s argmax strictly beats Z_{oracle} .
3. V1 changes the oracle score but still leaves $\Delta_{\text{joint}} > 0$; the future-direction feature is therefore insufficient rather than corrective. V2 is more severe: it drives the oracle score to ≈ 0 on **residual-spiky-shocks** and **mixed-tail-soft**, so the current–future cross product is not merely insufficient — it actively ranks Z_{oracle} below alternatives.

B.2 V4 (λ -sweep with the oracle subspace reward)

$S_4(Z) = S_{\text{HM}}(Z) + (\lambda/2)\|V_r^\top Z\|_F^2$. $\text{pa}^2[i] \in [0, 1]$ for $i \in \{0, 1\}$ are the two squared canonical cosines (sorted descending: $\text{pa}^2[0] \geq \text{pa}^2[1]$) between $\text{span}(\hat{Z}_{\text{joint}}^{(\sigma)})$ and $\text{span}(V_2)$; both equal 1 iff the two 2-planes coincide, $\text{pa}^2[1] = 0$ means one direction of $\hat{Z}_{\text{joint}}^{(\sigma)}$ is orthogonal to the oracle plane.

λ	matrix	oracle	joint	Δ_{joint}	$\text{pa}^2[0]$	$\text{pa}^2[1]$
0	diffuse-diffuse	0.0066	0.0383	+0.0316	0.45	0.30
0	residual-spiky-shocks	0.0270	0.0409	+0.0140	0.88	0.56
0	mixed-tail-soft	0.0215	0.0386	+0.0170	0.48	0.01
0.01	diffuse-diffuse	0.0294	0.0396	+0.0102	0.93	0.61
0.01	residual-spiky-shocks	0.0326	0.0429	+0.0103	0.94	0.71
0.01	mixed-tail-soft	0.0294	0.0405	+0.0112	0.65	0.50
0.05	diffuse-diffuse	0.0344	0.0439	+0.0095	0.90	0.81
0.05	residual-spiky-shocks	0.0389	0.0536	+0.0147	0.89	0.70
0.05	mixed-tail-soft	0.0307	0.0481	+0.0174	0.68	0.59
0.1	diffuse-diffuse	0.0407	0.0542	+0.0135	0.77	0.74
0.1	residual-spiky-shocks	0.0563	0.0633	+0.0070	0.99	0.85
0.1	mixed-tail-soft	0.0675	0.0695	+0.0020	0.99	0.93
0.5	diffuse-diffuse	0.1077	0.1125	+0.0048	0.97	0.93
0.5	residual-spiky-shocks	0.1775	0.1816	+0.0041	0.99	0.96
0.5	mixed-tail-soft	0.1856	0.1887	+0.0030	0.99	0.96

At $\lambda = 0$, $\hat{Z}_{\text{joint}}^{(\sigma)}$ on **mixed-tail-soft** has $\text{pa}^2[1] = 0.01$: its second direction is essentially *orthogonal* to $\text{span}(V_2)$. By $\lambda = 0.1$, the squared principal cosines on **residual-spiky-shocks** and **mixed-tail-soft** reach 0.93–0.99 and the score gap shrinks to +0.002–+0.007; **diffuse-diffuse** requires $\lambda \geq 0.5$. The residual gap at $\lambda = 0.5$ is structural: $\|V_r^\top Z\|_F^2$ is invariant under right-rotations of Z inside $\text{span}(V_r) \cap B_{\text{union}}$, so within that 2-plane the HM landscape retains its own argmax, generically distinct from the specific orthonormalised Z_{oracle} .